Compiler Construction

Lab 2

Submitted By:

Mahnoor Farrukh (52926)

BSCS-6

Task 1:

Task 1.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    string input;
    cout << "Enter sentence: ";
    getline(cin, input);

    string token = "";

    for(int i = 0; i < input.length(); i++) {
        if(input[i] != ' ')
            token += input[i];
        else {
            cout << "Token: " << token << endl;
            token = "";
        }
    }

    cout << "Token: " << token << endl;

    return 0;
}
```

C:\Users\mahno\OneDrive\Dc

```
Enter sentence: We love compiler
Token: We
Token: love
Token: compiler

--------------------------------
Process exited after 8.851 seconds with return value 0
Press any key to continue . . .
```

Task 2:

Task 1.cpp | Task2.cpp

```cpp
#include <iostream>
using namespace std;

int main() {

    string input;
    cout << "Enter statement: ";
    getline(cin, input);

    string token = "";

    for(int i = 0; i < input.length(); i++) {

        if(input[i] != ' ')
            token += input[i];

        else {

            if(token == "int")
                cout << token << " : KEYWORD" << endl;

            else if(token == "+" || token == "=")
                cout << token << " : OPERATOR" << endl;

            else if(token[0] >= '0' && token[0] <= '9')
                cout << token << " : NUMBER" << endl;

            else
                cout << token << " : IDENTIFIER" << endl;

            token = "";
        }
    }
```

C:\Users\mahno\OneDrive\Dc

```
Enter statement: int total = sum + 10
int : KEYWORD
total : IDENTIFIER
= : OPERATOR
sum : IDENTIFIER
+ : OPERATOR
10 : NUMBER

--------------------------------
Process exited after 12.33 seconds with return value
Press any key to continue . . .
```

```cpp
            else if(token[0] >= '0' && token[0] <= '9')
                cout << token << " : NUMBER" << endl;

            else
                cout << token << " : IDENTIFIER" << endl;

            token = "";
        }
    }

    // check last token
    if(token == "int")
        cout << token << " : KEYWORD" << endl;

    else if(token == "+" || token == "=")
        cout << token << " : OPERATOR" << endl;

    else if(token[0] >= '0' && token[0] <= '9')
        cout << token << " : NUMBER" << endl;

    else
        cout << token << " : IDENTIFIER" << endl;

    return 0;
```

Task 3:

(globals)

Task 1.cpp | Task2.cpp | Task3.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    string input;
    cout << "Enter statement: ";
    getline(cin, input);

    string token = "";

    for(int i = 0; i < input.length(); i++) {
        if(input[i] != ' ')
            token += input[i];
        else {
            if(token == "int" || token == "float" || token == "if" || token == "while" || token == "for")
                cout << token << " : KEYWORD" << endl;
            else if(token == "+" || token == "=")
                cout << token << " : OPERATOR" << endl;
            else if(token[0] >= '0' && token[0] <= '9')
                cout << token << " : NUMBER" << endl;
            else
                cout << token << " : IDENTIFIER" << endl;

            token = "";
        }
    }

    // Last token
    if(token == "int" || token == "float" || token == "if" || token == "while" || token == "for")
        cout << token << " : KEYWORD" << endl;
    else if(token == "+" || token == "=")
        cout << token << " : OPERATOR" << endl;
    else if(token[0] >= '0' && token[0] <= '9')
        cout << token << " : NUMBER" << endl;
    else
        cout << token << " : IDENTIFIER" << endl;

    return 0;
}
```

C:\Users\mahno\OneDrive\Dc

```
Enter statement: for i = 0
for : KEYWORD
i : IDENTIFIER
= : OPERATOR
0 : NUMBER

--------------------------------
Process exited after 5.6 seconds with retur
Press any key to continue . . .
```